Def. Let W be a vector field on $\mathbb{R}^3$, and v_p be a tangent vector. A covariant derivative of W with respect to v_p is the tangent vector

$$\nabla_v W \overset{\text{def}}{=} W(p+t\cdot v)'(0)_p$$

Lemma. If $W=\sum_{i=1}^{3} w_i\cdot e_i$ is a vector field on $\mathbb{R}^3$, then for any tangent vector v_p,

$$\nabla_v W = \sum_{i=1}^{3} v[w_i]\cdot e_i(p).$$

Proof. We have $W(p+t\cdot v)=\sum_{i=1}^{3} \underbrace{w_i(p+tv)}_{\text{scalar valued}}\cdot \underbrace{e_i(p+tv)}_{\text{vector valued}}$,

thus, by the definition of directional derivative,

$$\nabla_v W = \frac{d}{dt} W(p+tv)\Big|_{t=0} = \sum \frac{d}{dt} w_i(p+tv)\Big|_{t=0}\cdot e_i(p) = \sum v[w_i]\cdot e_i(p).$$